

Physiology of the Large Intestine

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Learning Outcome

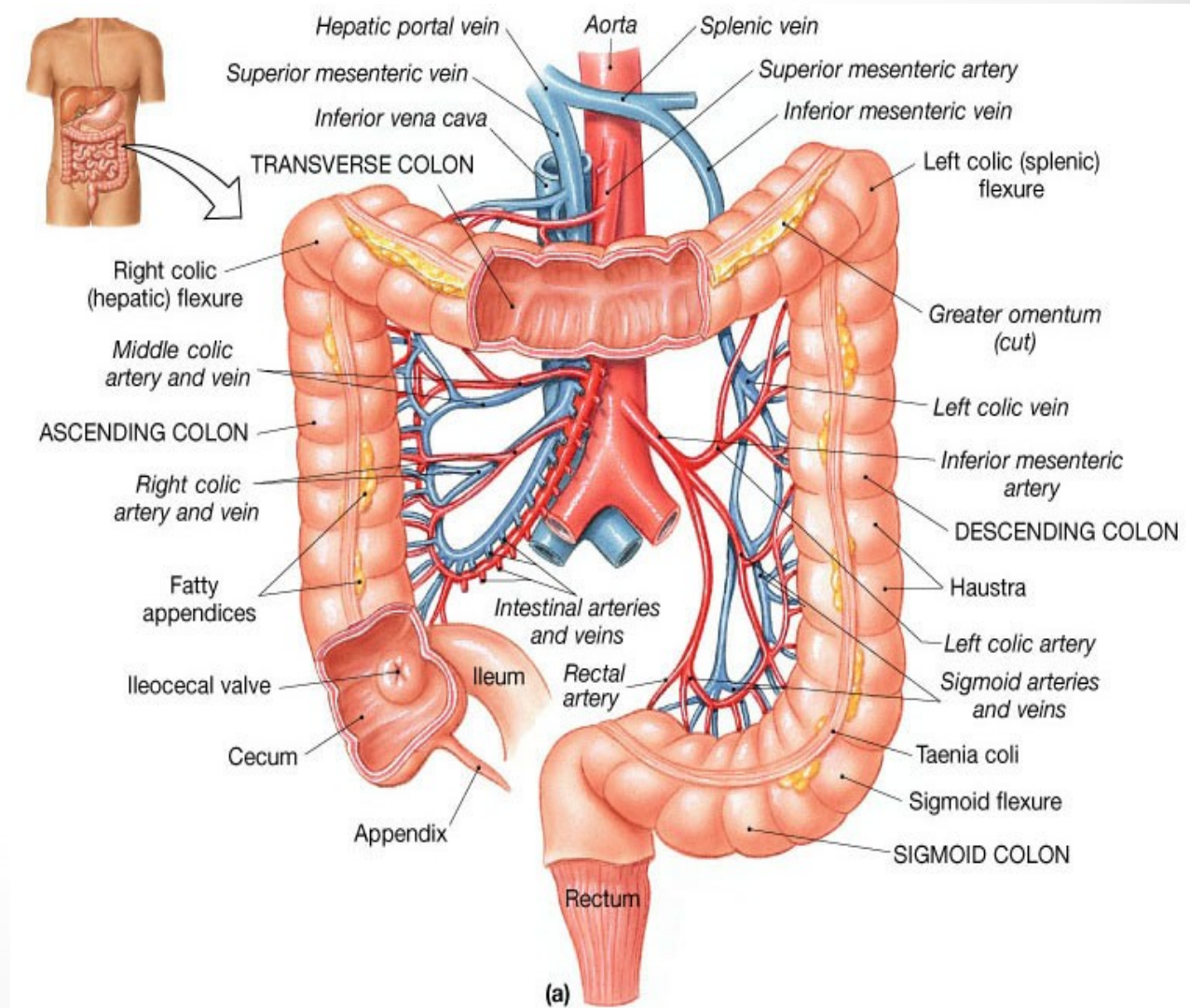
By the end of this lecture the students must be able to:

- Describe the main functions of different regions of large intestine.
- Describe the mechanism of these functions.

The areas of the colon are:

The large intestine is about 1.5 m long and 6.5 cm in diameter

1. Cecum with appendix
2. Ascending
3. Transverse
4. Descending
5. Sigmoid
6. Rectum
7. Anal canal



Functions of the Large Intestine

1. Haustral churning, peristalsis, and mass peristalsis drive contents of colon into rectum.
2. Bacteria in large intestine convert proteins to amino acids, break down amino acids, and produce some B vitamins like thiamine (B1), B12, folic acid and vitamin K. By this function, large intestine contributes in erythropoietic activity and blood clotting mechanism.

3. Absorption of some water, ions, vitamins, organic substances like glucose and alcohol with drugs like anesthetic agents, sedatives and steroids.

4. Formation of feces.

5. Defecation (emptying rectum).

6. Large intestine excretes heavy metals like mercury, lead, bismuth and arsenic through feces.

„7. Large intestine secretes mucin and inorganic substances like chlorides and bicarbonates.

Secretions of the Large Intestine

Mucus Secretion.

- The mucosa of the large intestine has many crypts of Lieberkühn.
- Absence of villi.
- The epithelial cells contain almost no enzymes.
- Mucosa Secrete mucus, Lubricates colon; protects mucosa.

Secretions of the Large Intestine

- Stimulation of the *pelvic nerves from the spinal* cord can cause marked increase in mucus secretion. This occurs along with increase in peristaltic motility of the colon.
- During extreme parasympathetic stimulation, so much mucus can be secreted into the large intestine that the person has a bowel movement of ropy mucus as often as every 30 minutes; this mucus often contains little or no fecal material.

Motility in the Large Intestine

- The proximal half of the colon is concerned with absorption and the distal half with storage
- The transit of small labeled markers through the large intestine occurs in 36-48 hrs
- Movements of the colon
 - Mixing movements (Haustrations)
 - Propulsive movements (Mass Movements)

Mechanical Digestion in the Large Intestine

- The passage of chyme from the ileum into the cecum is regulated by the action of the ileocecal sphincter. Normally, the valve remains partially closed so that the passage of chyme into the cecum usually occurs slowly. Immediately after a meal, a gastroileal reflex intensifies peristalsis in the ileum and forces any chyme into the cecum. The hormone gastrin also relaxes the sphincter. Whenever the cecum is distended, the degree of contraction of the ileocecal sphincter intensifies.

- Movements of the colon begin when substances pass the ileocecal sphincter. Because chyme moves through the small intestine at a fairly constant rate, the time required for a meal to pass into the colon is determined by gastric emptying time. As food passes through the ileocecal sphincter, it fills the cecum and accumulates in the ascending colon.

- One movement characteristic of the large intestine is haustral churning. In this process, the haustra remain relaxed and become distended while they fill up when the distension reaches a certain point, the walls contract and squeeze the contents into the next haustrum. Peristalsis also occurs, although at a slower rate (3–12 contractions per minute) than in more proximal portions of the tract.

- A final type of movement is mass peristalsis, a strong peristaltic wave that begins at about the middle of the transverse colon and quickly drives the contents of the colon into the rectum. Because food in the stomach initiates this gastrocolic reflex in the colon, mass peristalsis usually takes place three or four times a day, during or immediately after a meal.

Chemical Digestion in the Large Intestine

- The final stage of digestion occurs in the colon through the activity of bacteria that inhabit the lumen. Mucus is secreted by the glands of the large intestine, but no enzymes are secreted. Chyme is prepared for elimination by the action of bacteria, which ferment any remaining carbohydrates and release hydrogen, carbon dioxide, and methane gases. These gases contribute to flatus (gas) in the colon, termed flatulence when it is excessive.

- Bacteria also convert any remaining proteins to amino acids and break down the amino acids into simpler substances: indole, skatole, hydrogen sulfide, and fatty acids. Some of the indole and skatole is eliminated in the feces and contributes to their odor; the rest is absorbed and transported to the liver, where these compounds are converted to less toxic compounds and excreted in the urine. Bacteria also decompose bilirubin to simpler pigments, including stercobilin, which gives feces their brown color. Bacterial products that are absorbed in the colon include several vitamins needed for normal metabolism, among them some B vitamins and vitamin K

Absorption and Feces Formation in the Large Intestine

- Most of the absorption in the large intestine occurs in the proximal one half of the colon, giving this portion the name *absorbing colon*, whereas the distal colon functions principally for feces storage until a propitious time for feces excretion and is therefore called the *storage colon*

Absorption and Feces Formation in the Large Intestine

- By the time chyme has remained in the large intestine 3–10 hours, it has become solid or semisolid because of water absorption and is now called feces. Chemically, feces consist of water, inorganic salts, sloughed-off epithelial cells from the mucosa of the gastrointestinal tract, bacteria, products of bacterial decomposition, unabsorbed digested materials, and indigestible parts of food. Although 90% of all water absorption occurs in the small intestine, the large intestine absorbs enough to make it an important organ in maintaining the body's water balance. Of the 0.5–1.0 liter of water that enters the large intestine, all but about 100–200 mL is normally absorbed via osmosis. The large intestine also absorbs ions, including sodium and chloride, and some vitamins.

- Reabsorption in the large intestine includes:
 - Water
 - Vitamins – K, biotin (B7), and B₅
 - Organic wastes – urobilinogens and sterobilinogens
 - Bile salts
 - Toxins

The Rectum

- Last portion of the digestive tract
- Terminates at the anal canal
- Internal and external anal sphincters